Advanced Topics in Networking

CSE 322

At A Glance …

- An advanced execution
- An advanced command
- An advanced tool

An Advanced Execution

- TCP Congestion Control Algorithms (CCAs) in Linux
  - Source: https://jasonmurray.org/posts/2021/tcpcca/

- Check the current CCA
  - sysctl net.ipv4.tcp_congestion_control

Output Example:

```
jemurray@home-server:~$ sysctl net.ipv4.tcp_congestion_control
net.ipv4.tcp_congestion_control = cubic
```

TCP CCA in Linux

- Check for the default CCAs in Ubuntu
  - sysctl net.ipv4.tcp_available_congestion_control

Output Example:

```
jemurray@home-server:~$ sysctl net.ipv4.tcp_available_congestion_control
net.ipv4.tcp_available_congestion_control = reno cubic
```

TCP CCA in Linux (contd.)

- List installed, but not necessarily loaded CCAs
  - ls -al /lib/modules/`uname -r`/kernel/net/ipv4/tcp*

Output Example:

```
rver:~$ ls -al /lib/modules/`uname -r`/kernel/net/ipv4/tcp*
t root 14113 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_bbr.ko
t root 10913 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_bic.ko
t root 13865 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_cdg.ko
t root 10041 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_dctcp.ko
t root  8401 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_diag.ko
t root  7489 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_highspeed.ko
t root  9985 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_htcp.ko
t root  8073 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_hybla.ko
t root  8985 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_illinois.ko
t root  8617 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_lp.ko
t root 12561 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_nv.ko
t root 13441 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_probe.ko
t root  6449 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_scalable.ko
t root 11073 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_vegas.ko
t root  7609 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_veno.ko
t root  8097 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_westwood.ko
t root  7945 May  7 19:44 /lib/modules/4.15.0-144-generic/kernel/net/ipv4/tcp_yeah.ko
```

TCP CCA in Linux (contd.)

- Use modprobe to load additional modules
  - sudo /sbin/modprobe tcp_bbr

Output Example:

```
jemurray@home-server:~$ sudo /sbin/modprobe tcp_bbr

jemurray@home-server:~$ sysctl net.ipv4.tcp_available_congestion_control
net.ipv4.tcp_available_congestion_control = reno cubic bbr
```

TCP CCA in Linux (contd.)

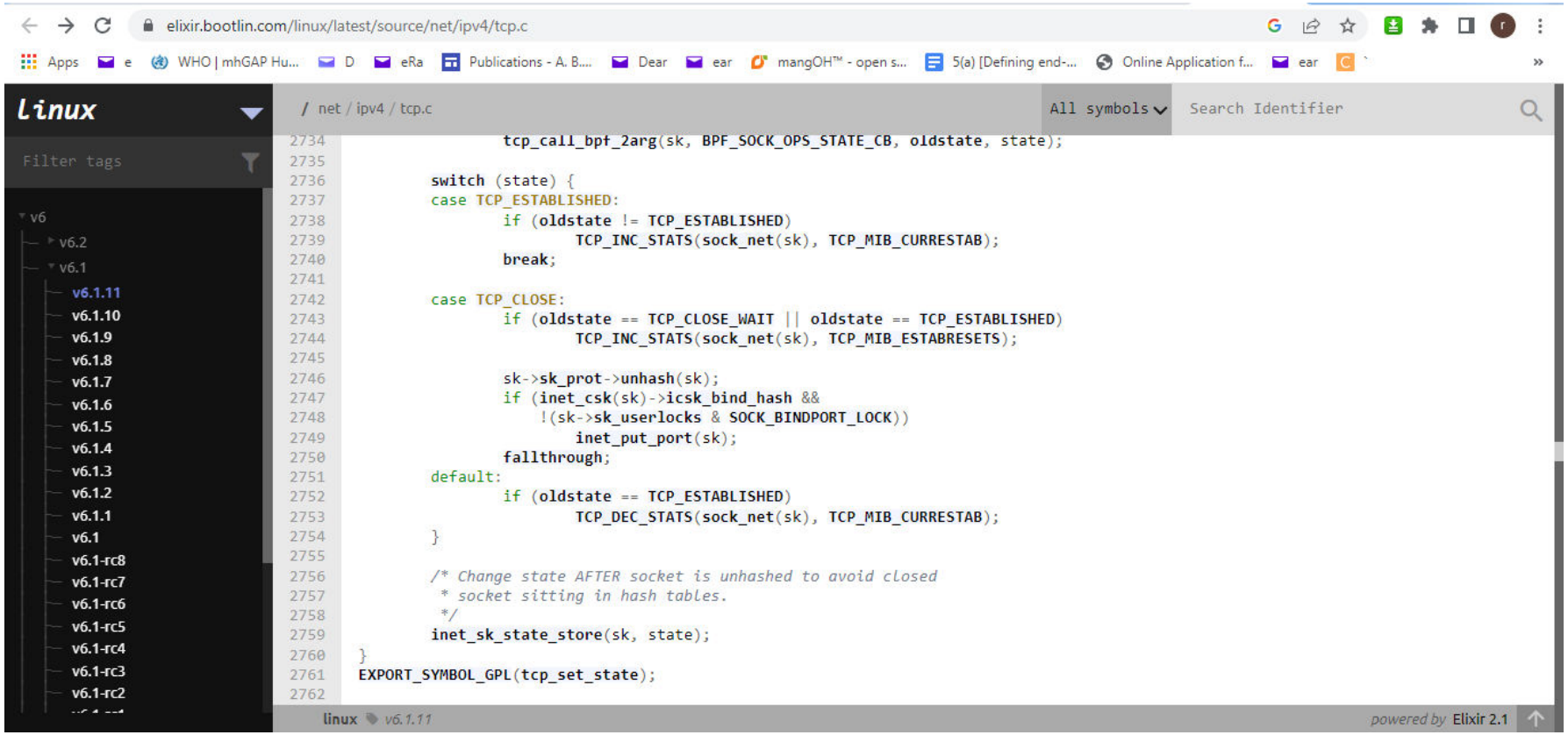
- Use sysctl to temporarily change the CCA to bbr
  - sudo /sbin/sysctl –w net.ipv4.tcp_congestion_control=bbr

```
jemurray@home-server:~$ sudo /sbin/sysctl -w net.ipv4.tcp_congestion_control=bbr
net.ipv4.tcp_congestion_control = bbr

jemurray@home-server:~$ sysctl net.ipv4.tcp_congestion_control
net.ipv4.tcp_congestion_control = bbr
```

TCP Code in Ubuntu

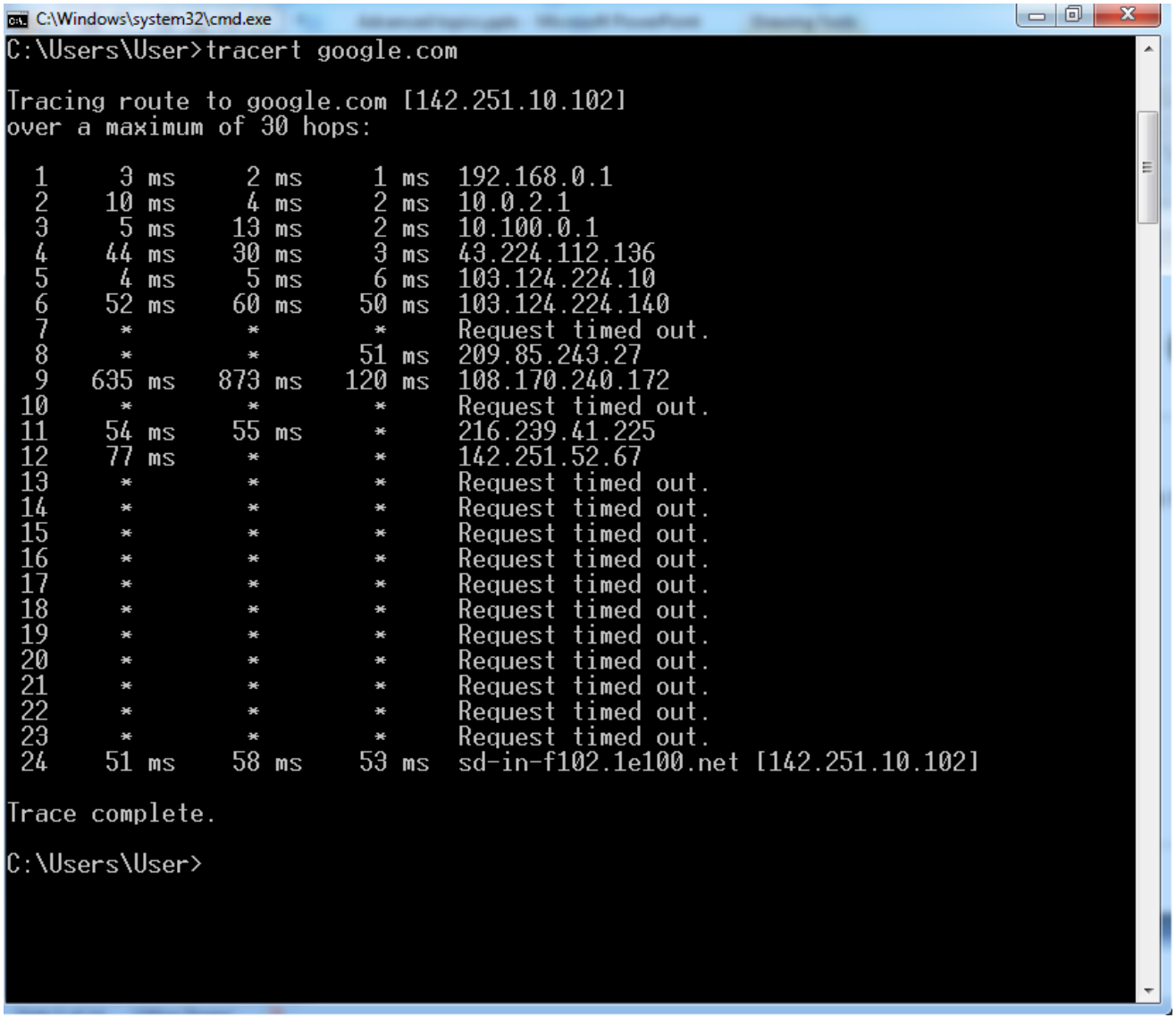
- Tentative location: /net/ipv4/tcp.c
- A snapshot of a part of the code …



An Advanced Command

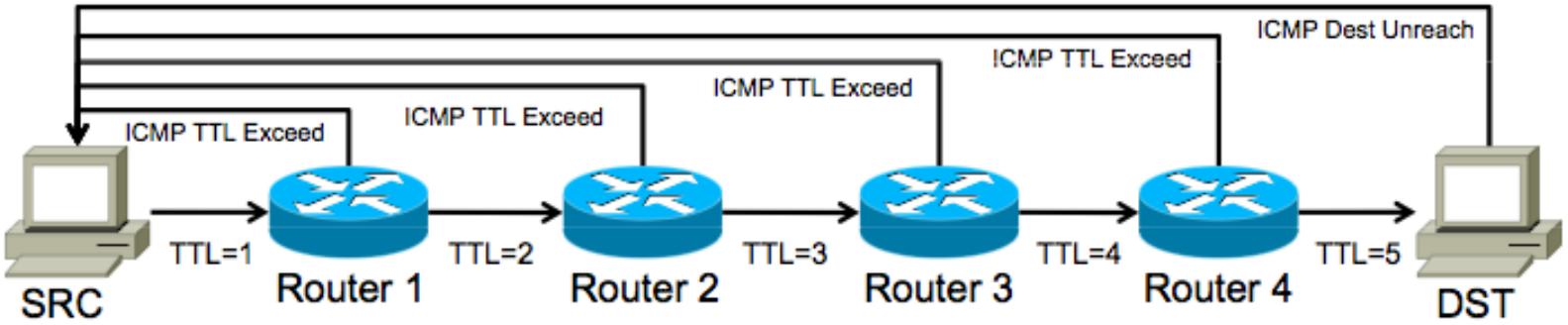
- traceroute
  - Determines the route to a node
    - tracert: Windows
    - traceroute: Unix

A Sample Output of “tracert”



How Does Traceroute Work?

1. Launch a probe packet towards DST, with a TTL of 1
2. Every router hop decrements the IP TTL of the packet by 1
3. When the TTL hits 0, packet is dropped, router sends ICMP TTL Exceed packet to SRC with the original probe packet as payload
4. SRC receives this ICMP message, displays a traceroute “hop”
5. Repeat from step 1, with TTL incremented by 1 each time, until…
6. DST host receives probe, returns ICMP Dest Unreachable

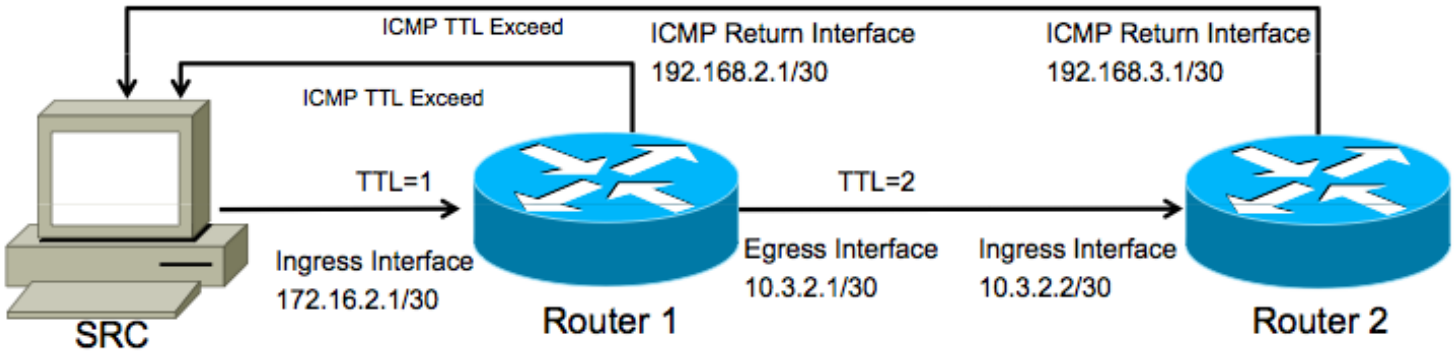


Responses in Traceroute

- All hops except the last one will (or should) return a "TTL exceeded in transit" message
- The last hop should return a "destination unreachable/port unreachable" message indicating that it cannot handle the received traffic
  - UDP traceroute packets are typically addressed to a pseudorandom high port on which the end host is not likely to be listening

Traceroute Report Hop

- Traceroute packet with TTL of 1 enters router via the ingress interface
- Router decrements TTL to 0, drops packet, generates ICMP TTL Exceed
  - ICMP packet dst address is set to the original traceroute probe source (SRC)
  - ICMP packet src address is set to the IP of the ingress router interface
  - Traceroute shows a result based on the src address of the ICMP packet
  - The above traceroute will read: 172.16.2.1 10.3.2.2
  - You have NO visibility into the return path or the egress interface used



Traceroute Latency Calculation

- How is traceroute latency calculated?
  - Timestamp when the probe packet is launched
  - Timestamp when the ICMP response is received
  - Calculate the difference to determine round-trip time
  - Routers along the path do not do anytime "processing"
    - They simply reflect the original packet's data back to the SRC
    - Many implementations encode the original launch timestamp into the probe packet, to increase accuracy and reduce state
  - Most Importantly: only the ROUNDTRIP is measured
    - Traceroute is showing you the hops on the forward path
    - But showing you latency based on the forward **PLUS** reverse path
      - Any delays on the reverse path will affect your results

An Advanced Tool

- Nmap
  - Source: https://nmap.org/download.html#windows
  - Extremely powerful
  - Extremely invasive
  - Extremely obvious if you are not careful
  - Extremely illegal if not done correctly

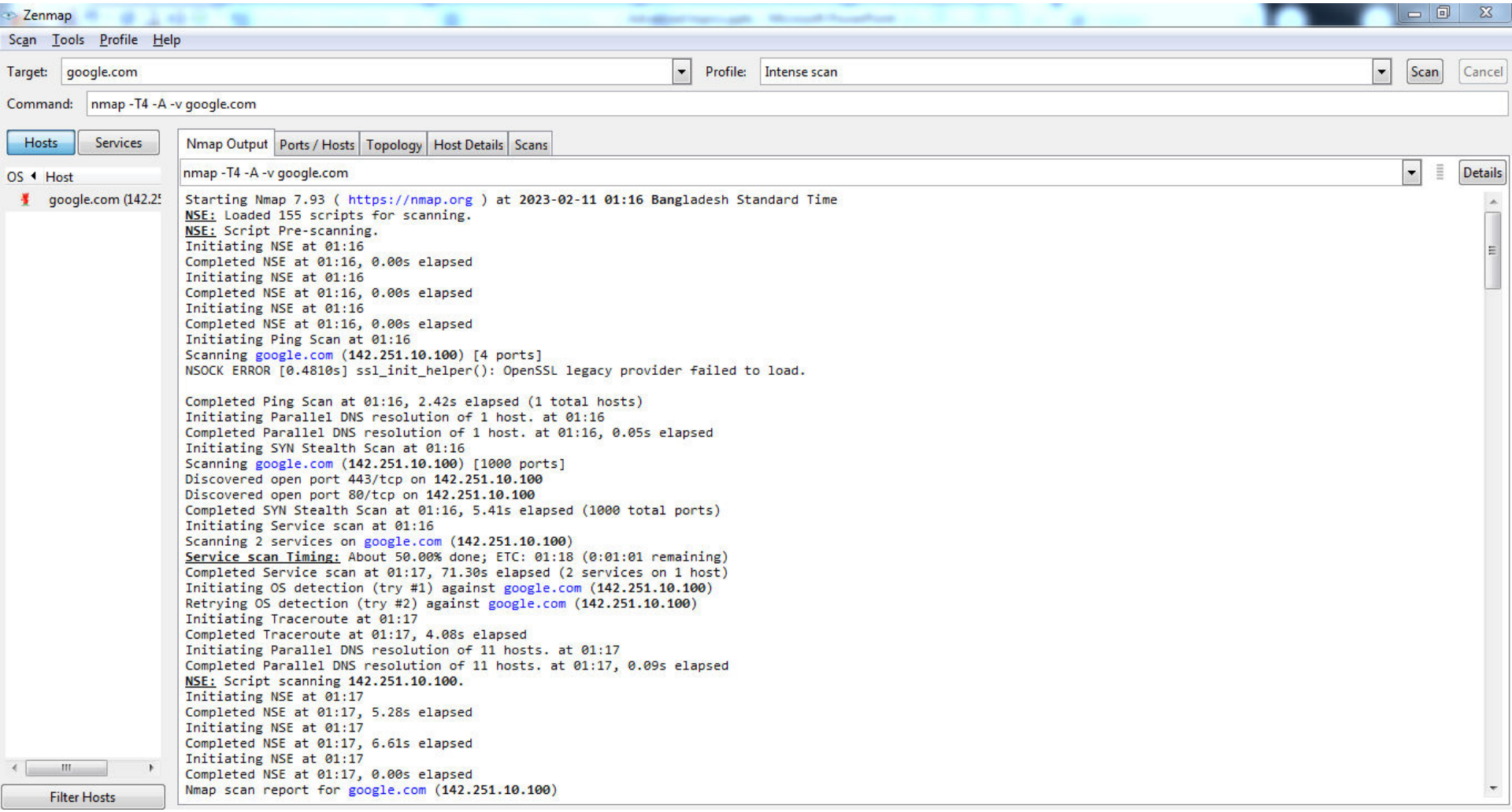
Port Scanning Basics

- nmap scans more than 1660 ports
- Most port scanners list ports as opened or closed
- nmap recognizes 6 port states
  - Open
    - Accepting TCP connections or UDP packets
  - Closed
    - Host is up on the IP address
    - Accessible but no app is listening
    - Try later

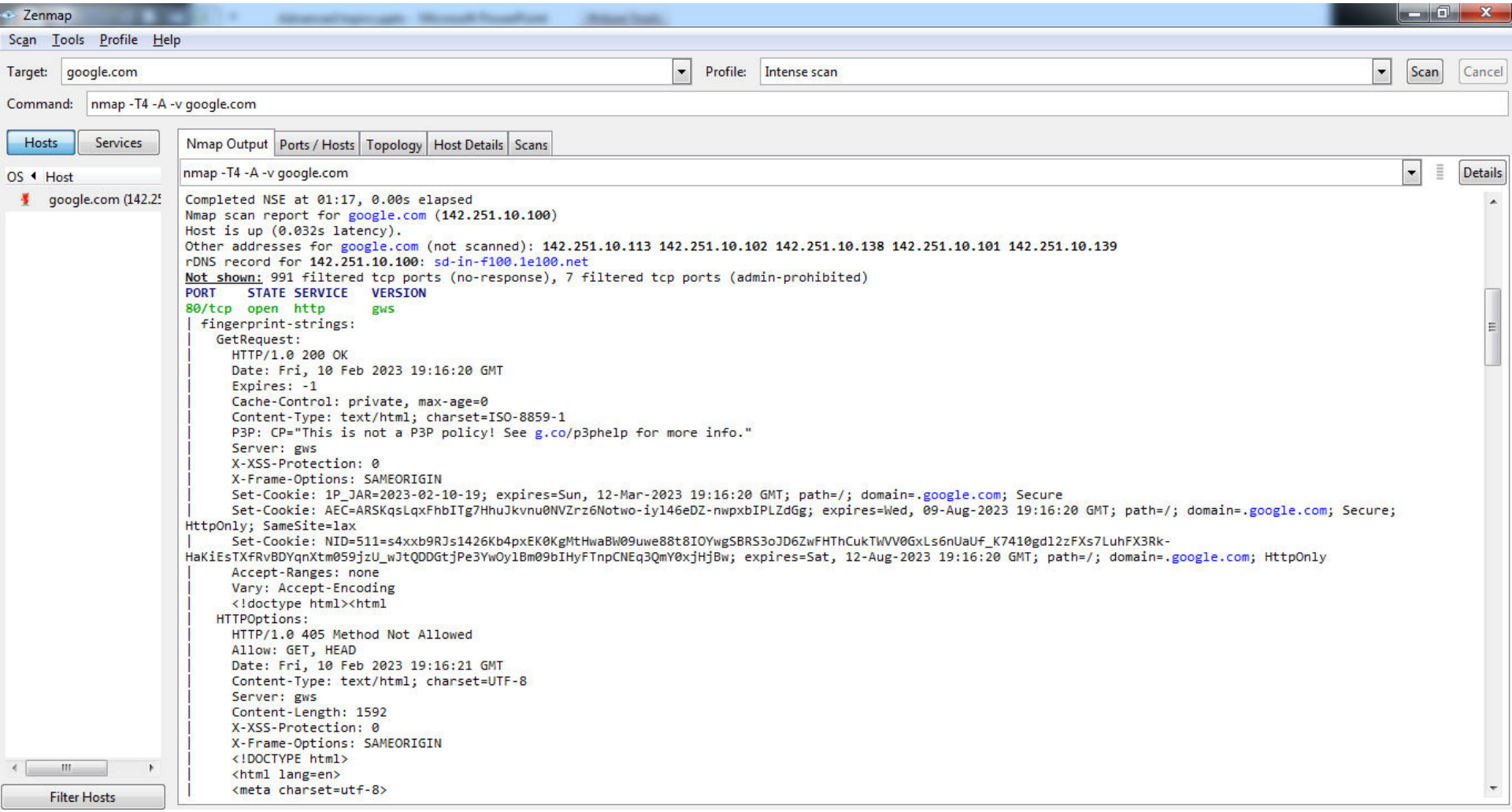
Port Scanning Basics (contd.)

- nmap recognizes 6 port states (contd.)
  - Filtered
    - No response from probe
      - Firewall probably did a stealth drop
    - Forces nmap to retry many times
  - Unfiltered
    - Port is accessible but cannot determine whether open or closed
  - Open|Filtered
    - When unable to determine whether port is open of filtered
  - Closed|Filtered
    - When unable to determine whether port is closed or filtered

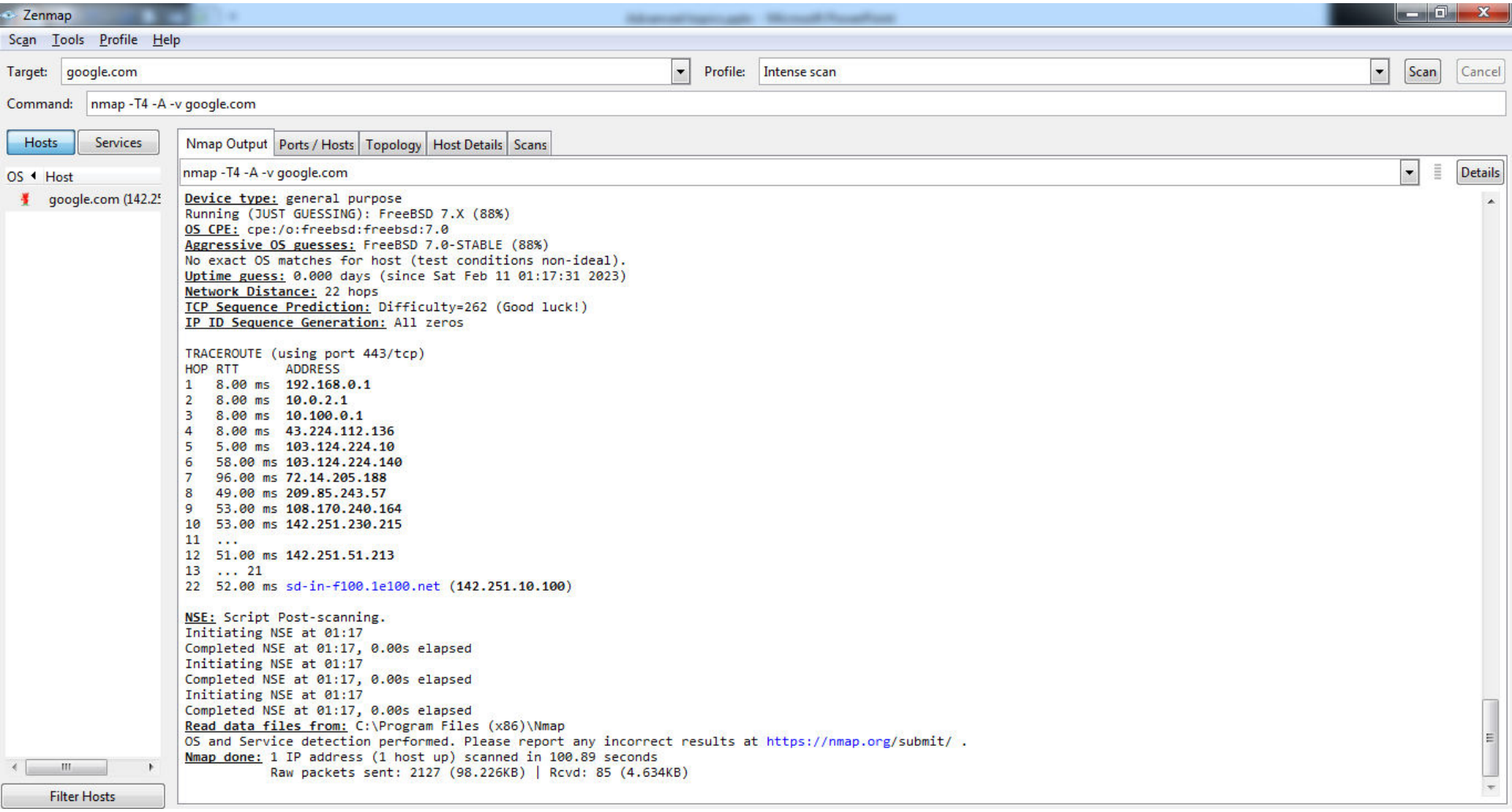
An Intense Scan in nmap: “Nmap Output”



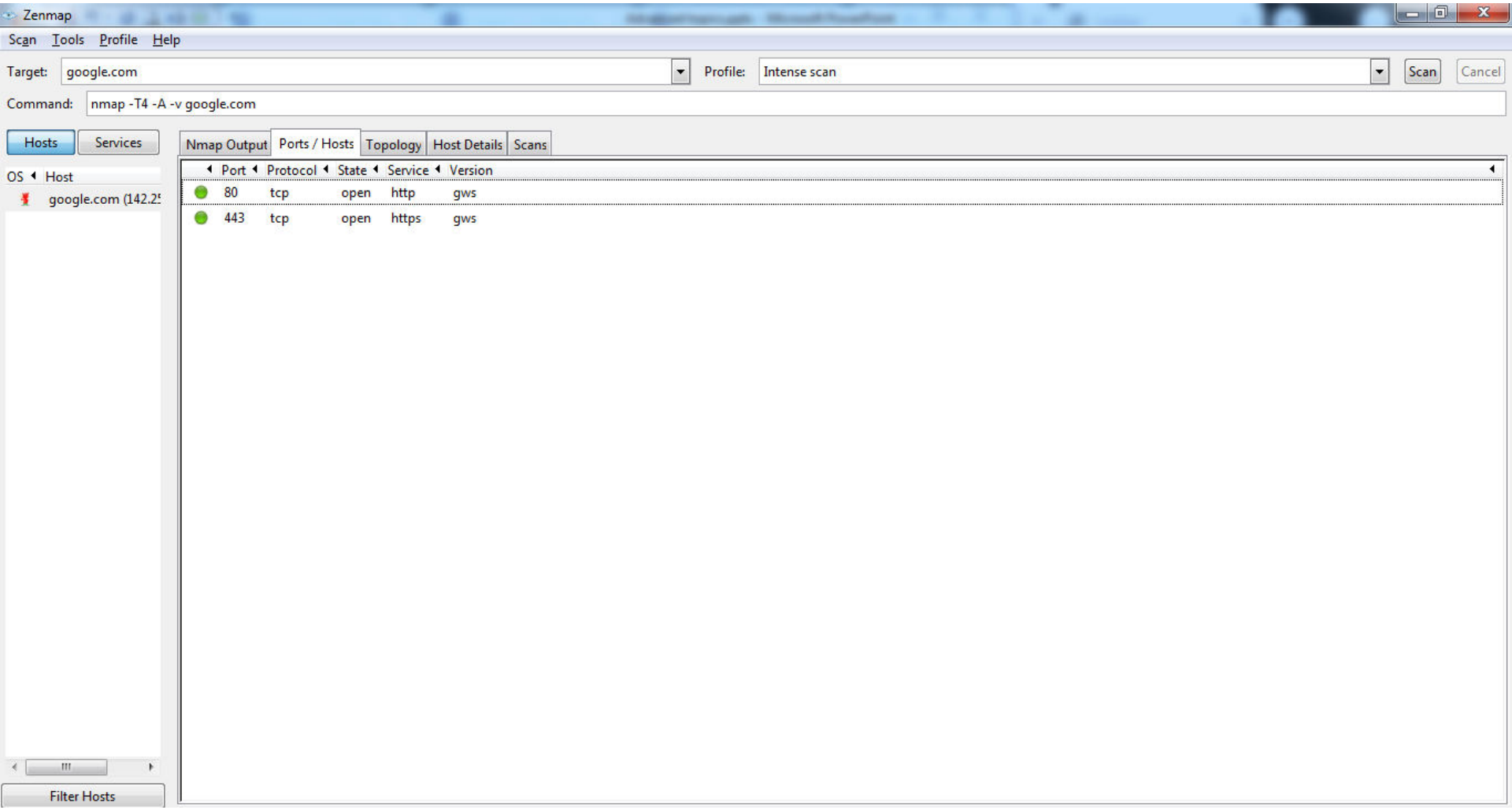
An Intense Scan in nmap : “Nmap Output” (contd.)



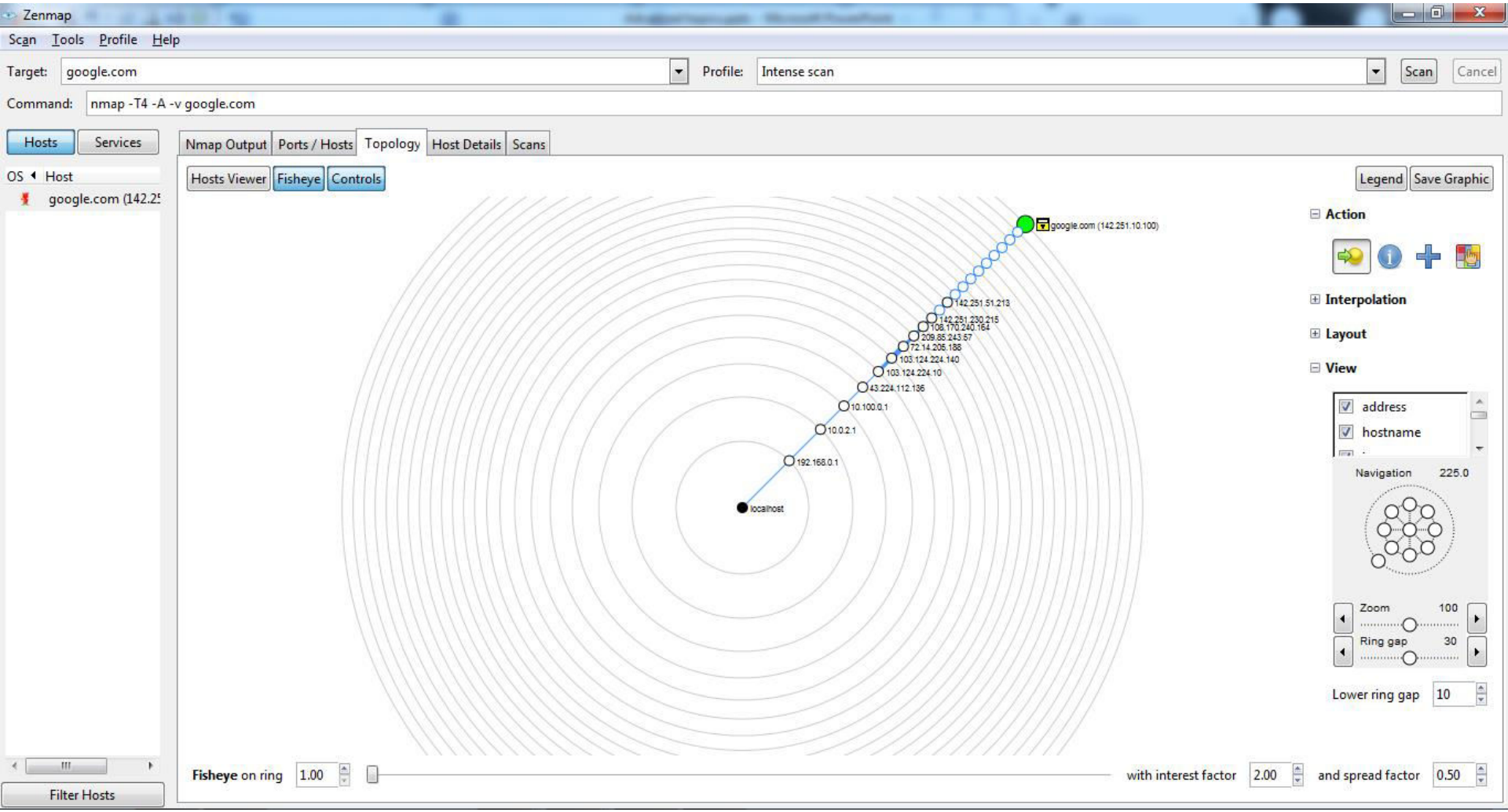
An Intense Scan in nmap : “Nmap Output” (contd.)



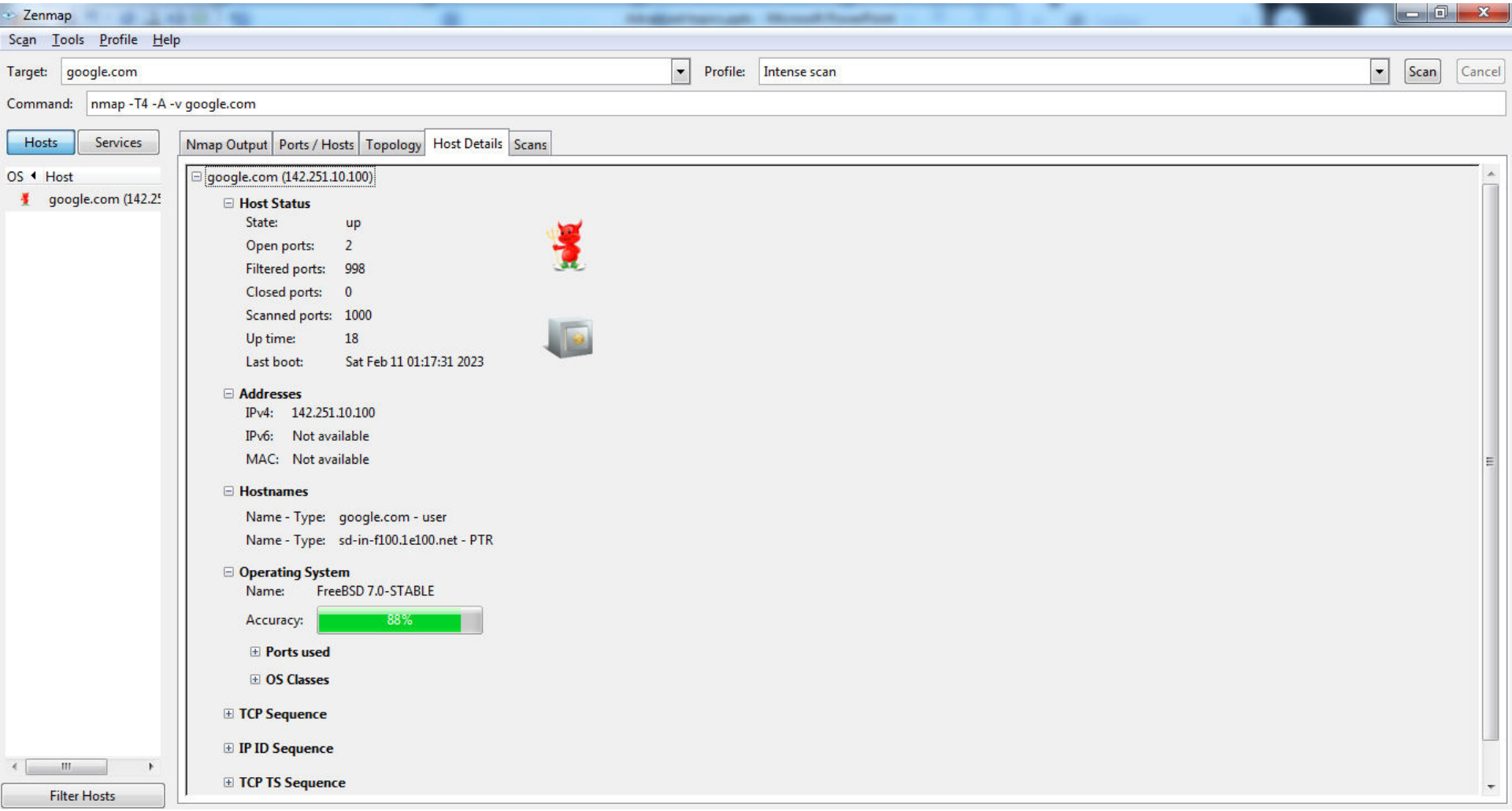
An Intense Scan in nmap: “Ports/Hosts”



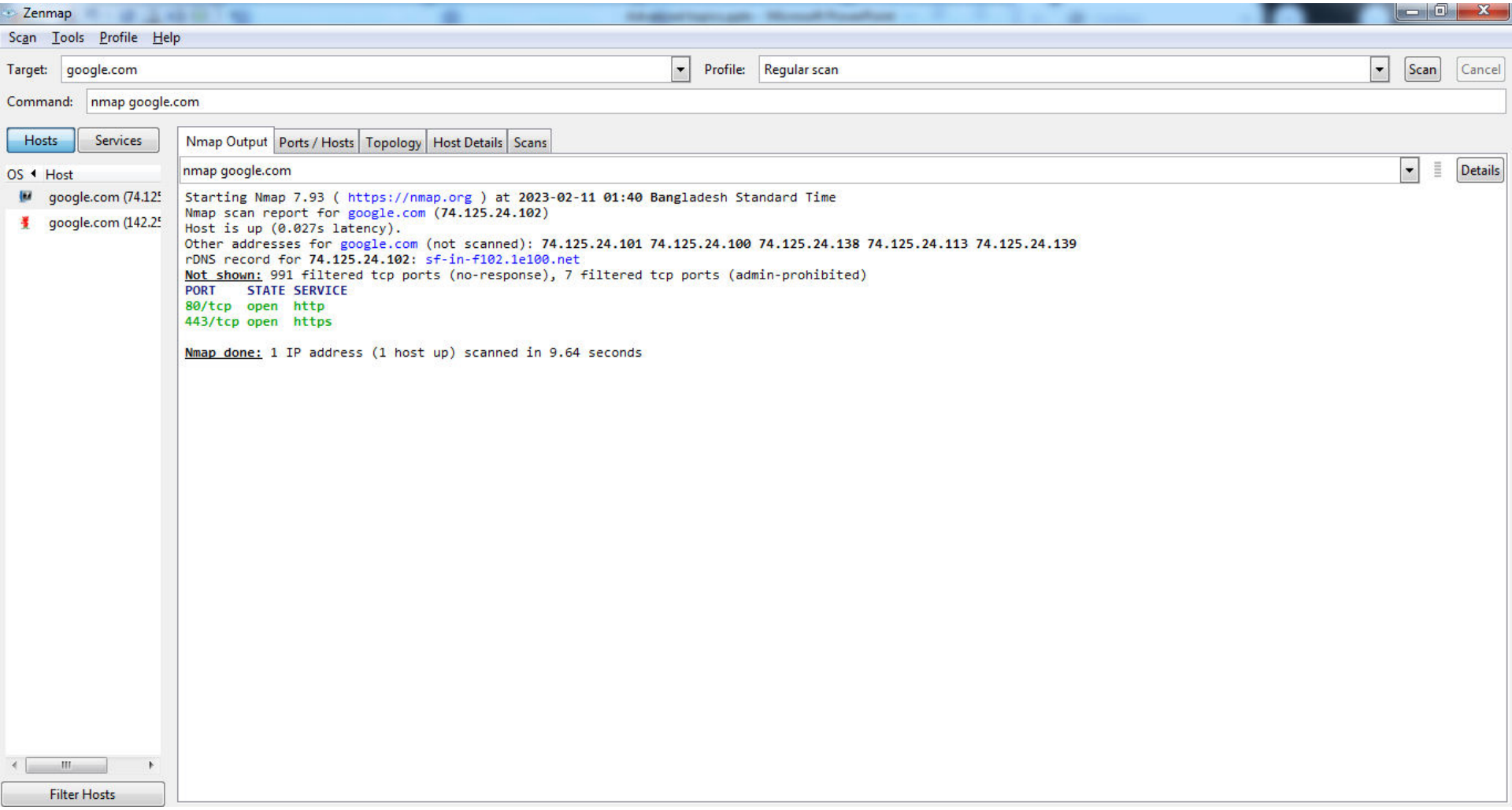
An Intense Scan in nmap: “Topology”



An Intense Scan in nmap: “Host Details”



A Regular Scan in nmap: “Nmap Output”



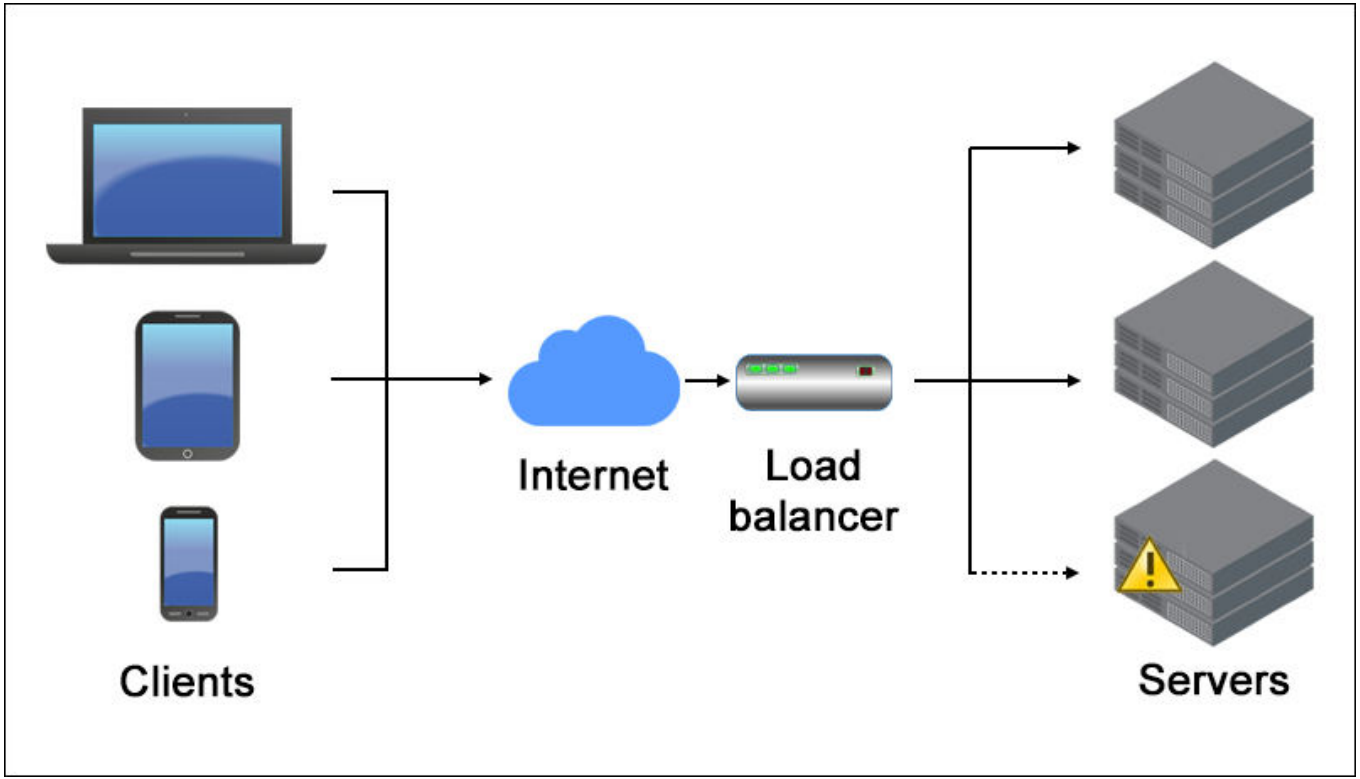
Thank You

Advanced Topics - 2

(Ack: Ishrat Jahan)

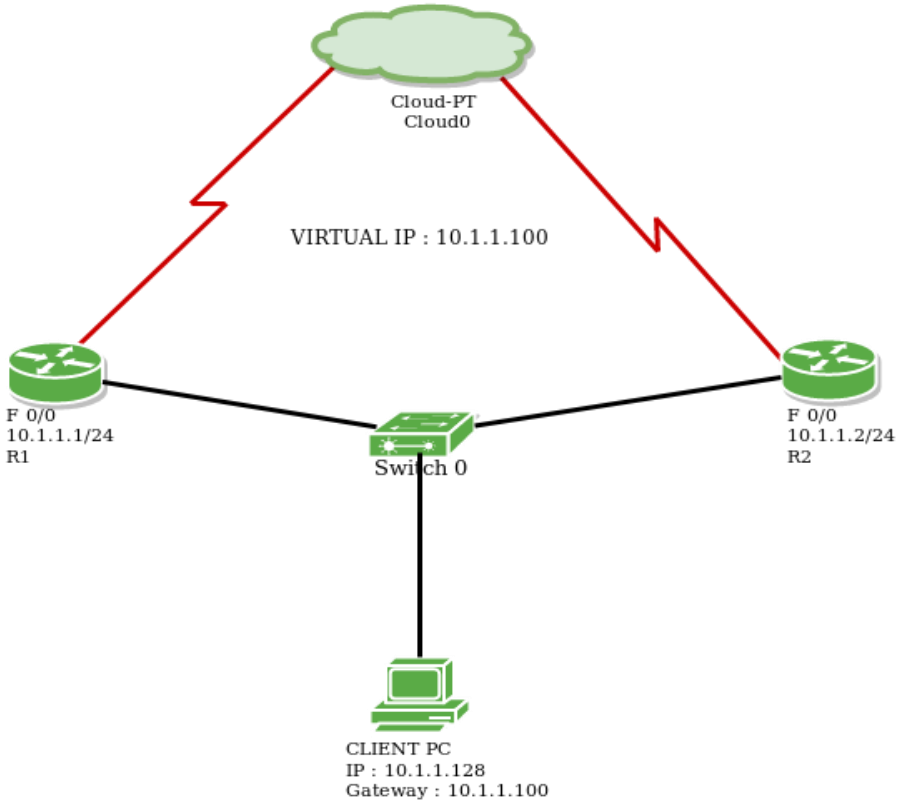
Load Balancing

Load Balancing

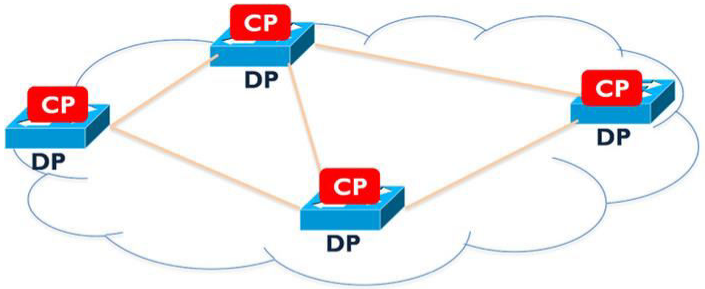


High Availability

VRRP - Virtual Router Redundancy Protocol

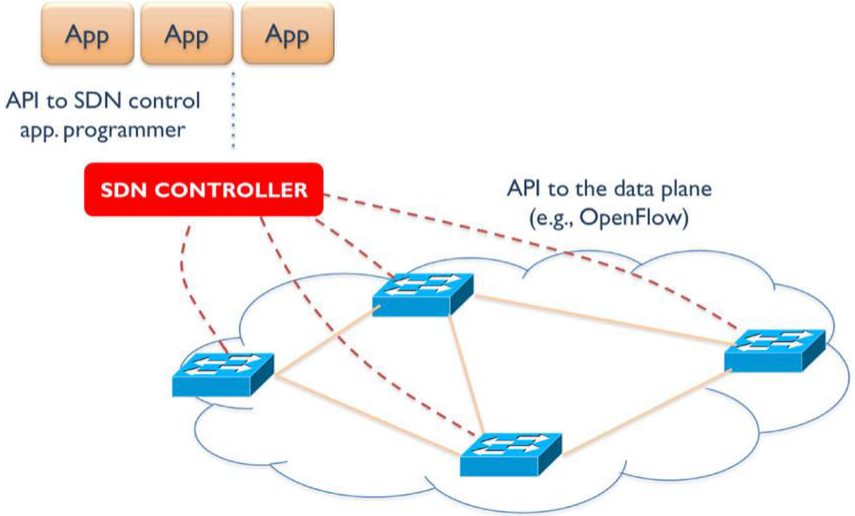


SDN (Software Defined Networking)



(a) Traditional networking

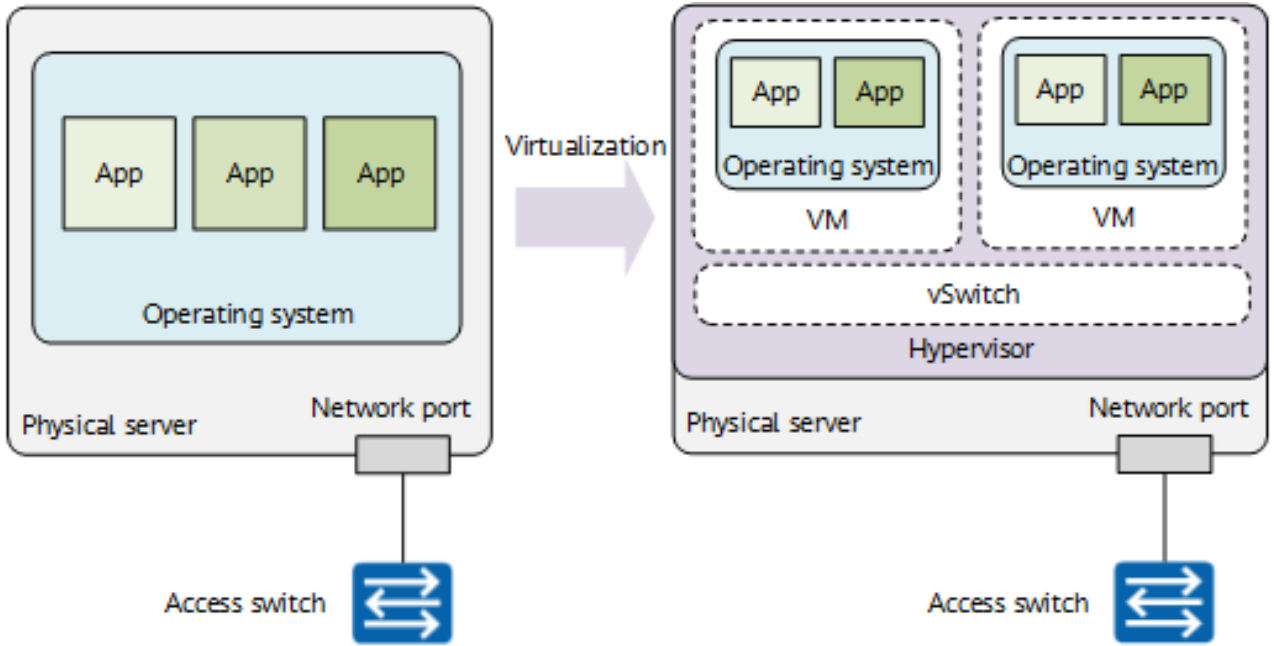
CP - Control Plane
DP - Data Plane



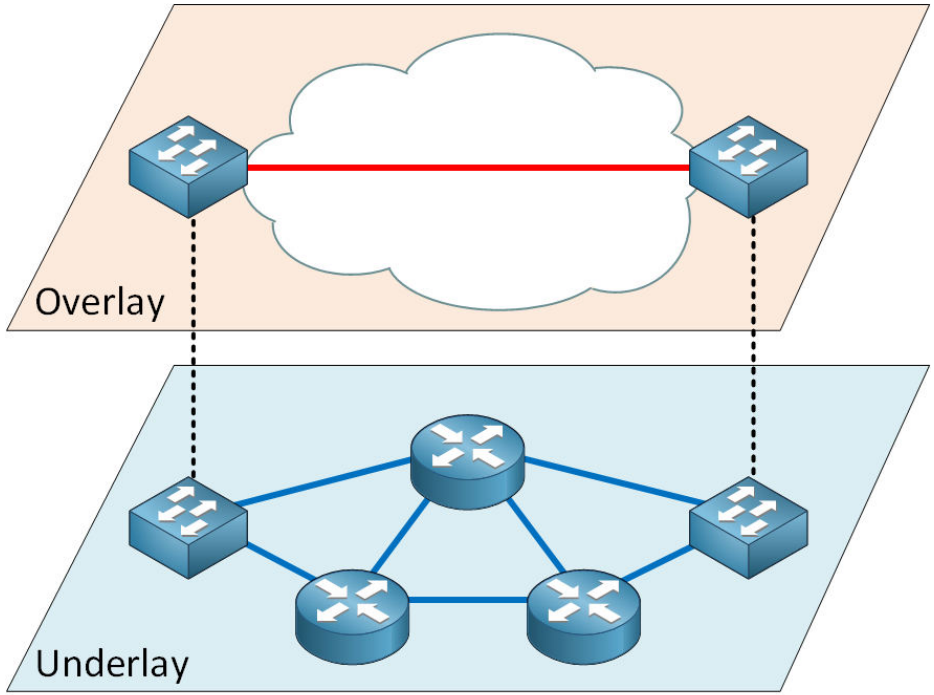
(b) SDN

Network Virtualization

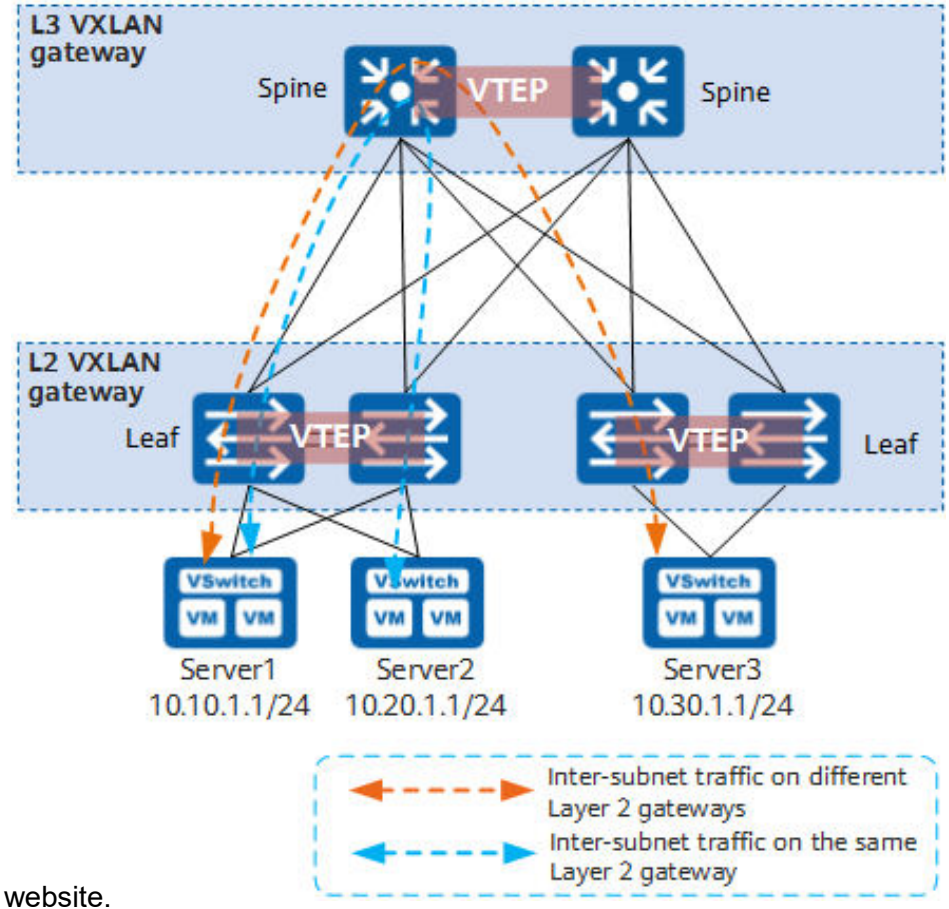
Server Virtualization



Network Virtualization (Overlay vs Underlay)



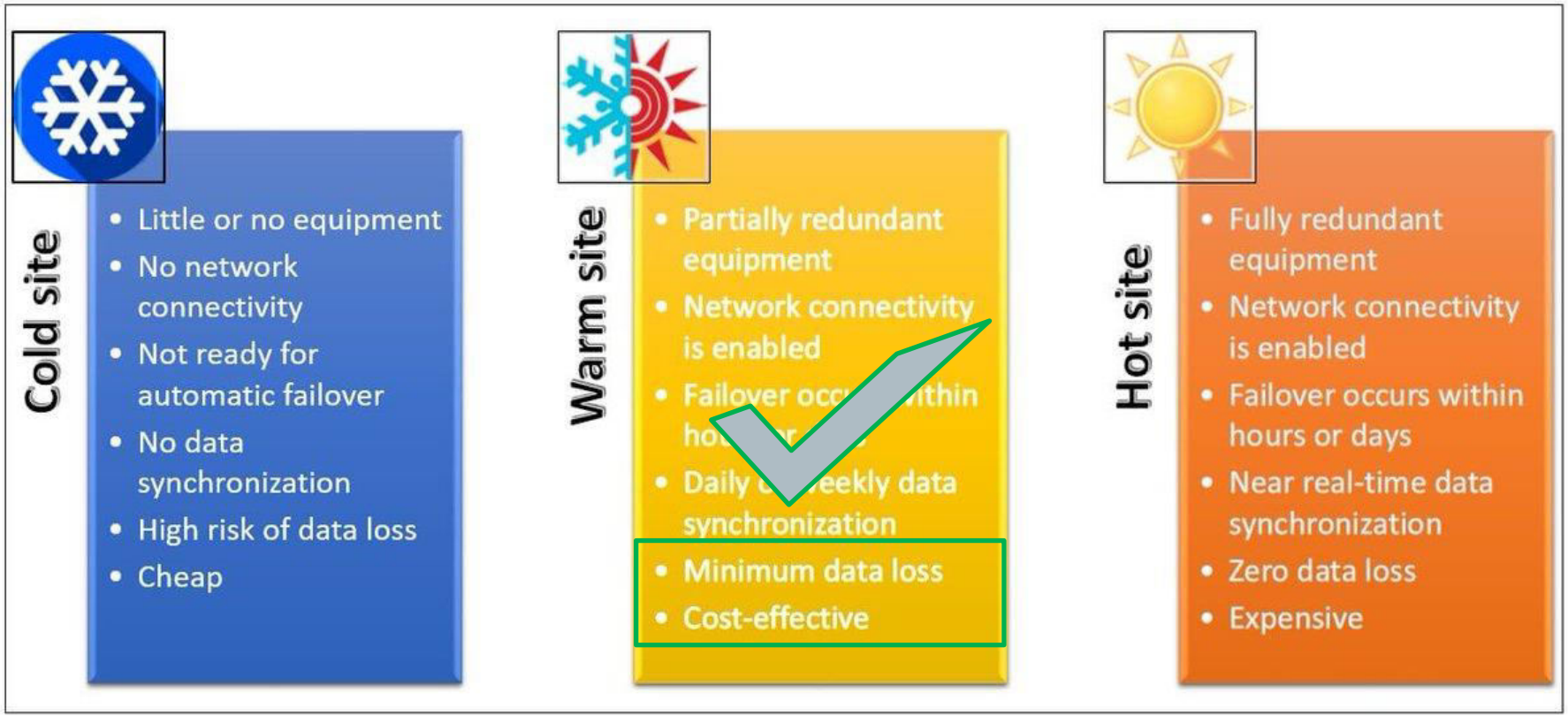
VXLAN



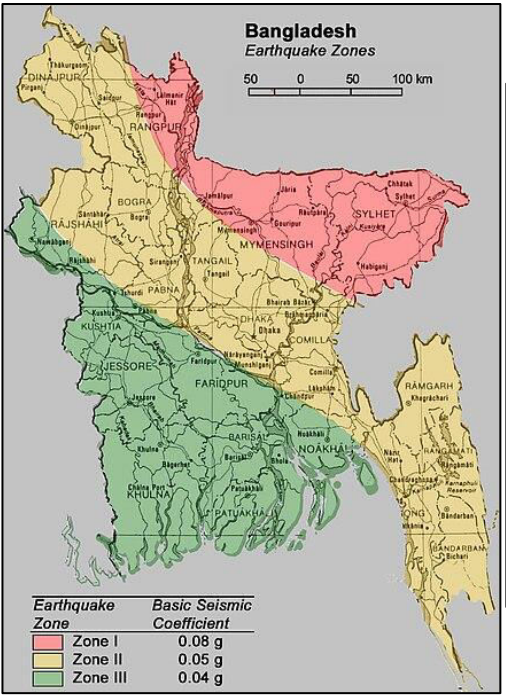
For details, please check this website.

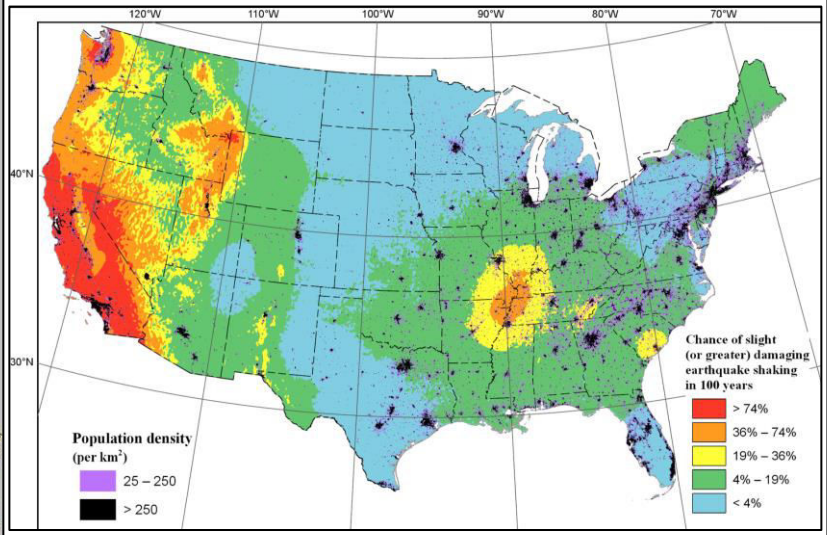
Data Centers and Disaster Recovery

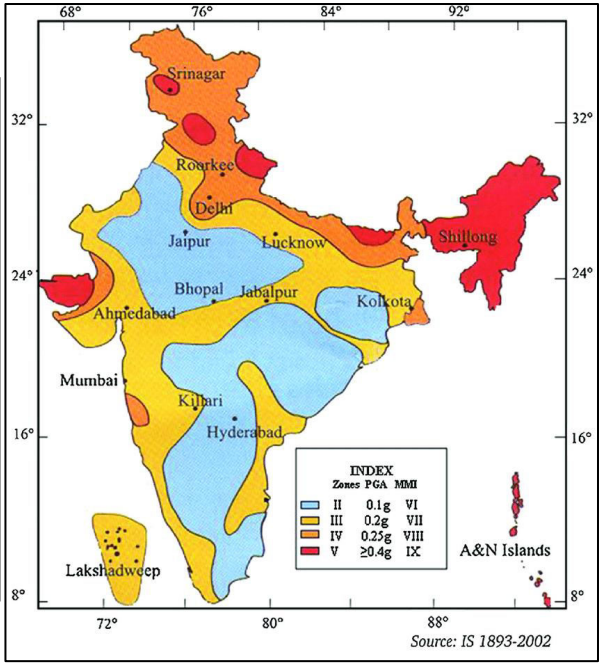
DC-DR


Cold site
Little or no equipment
No network connectivity
Not ready for automatic failover
No data synchronization
High risk of data loss
Cheap
Warm site
Partially redundant equipment
Network connectivity is enabled
Minimum data loss
Cost-effective
Hot site
Fully redundant equipment
Network connectivity is enabled
Failover occurs within hours or days
Near real-time data synchronization
Zero data loss
Expensive

Earthquake Zones – An Important Consideration for DR







Thank You!